

Solve volume problems

Adult-led preparation before the main lesson. Use precise vocabulary and one step at a time.

Name: _____ Date: _____

1. Which prior skill will help with today's lesson? Recognise, estimate and calculate the volume of cubes and cuboids.

2. A solid has 5 layers of 24 unit cubes. What is its volume?

3. Explain the first step you would take for: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?

4. Miro says: Miro calculates the surface area or adds the three dimensions. What should Miro check?

Teacher answers and checking prompts

1. Which prior skill will help with today's lesson? Recognise, estimate and calculate the volume of cubes and cuboids.
Answer Pupil identifies a relevant fact, representation or method.
2. A solid has 5 layers of 24 unit cubes. What is its volume?
Answer 120 cubic units.
3. Explain the first step you would take for: A box is 40 cm by 25 cm by 30 cm. How many 5 cm cubes fit exactly by volume?
Answer Identify the three perpendicular dimensions in one unit.
4. Miro says: Miro calculates the surface area or adds the three dimensions. What should Miro check?
Answer Volume measures space inside a solid, so multiply all three perpendicular dimensions and use cubic units.